

and regulated by IC 7805 (IC2). A 9V rechargeable battery is used for battery backup. The battery is charged through diode D6. Diode D7 is reverse-biased when the supply is present. The battery powers regulator 7805 to provide regulated 5V supply for the circuit in absence of mains supply. Capacitor C6 bypasses the ripples, if any, in the regulated supply. LED1 acts as the power indicator and resistor R10 limits the current through LED1.

Software

The software program is written in Assembly language and assembled using cross-assembler ASM51. Burn the generated Intel hex code into the microcontroller using a suitable programmer. The software works as per the flow-chart shown in Fig. 2.

When the microcontroller starts on pressing reset switch S1, it initialises the stack pointer and the LCD to display the message. The stack pointer, as a part of the RAM, stores the return address of the main program from subroutines. To initialise the LCD, the microcontroller sends a set of commands to the LCD. The `lcd_initialize` subroutine in the software is used to initialise the LCD. The delays for the LCD are provided by nested loops instead of the timer. These delays are used to provide timing delays at various locations of the software. The values for the loops are calculated according to the crystal frequency and the machine cycles taken by the respective instructions.

Initially, set the current time using

PARTS LIST

Semiconductors:

IC1	- AT89C51 microcontroller
IC2	- 7805, 5V regulator
T1	- BC547 npn transistor
D1-D7	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all ¼-watt, ±5% carbon, unless specified):

R1-R8	- 10-kilo-ohm
R9	- 2.2-kilo-ohm
R10	- 330-ohm
R11	- 15-ohm, 3-watt
RNW1	- 10-kilo-ohm resistor network
VR1	- 10-kilo-ohm preset

Capacitors:

C1, C2	- 33pF ceramic
C3	- 10µF, 16V electrolytic
C4	- 1000µF, 35V electrolytic
C5	- 330µF, 25V electrolytic
C6	- 0.1µF ceramic

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
S1-S5	- Push-to-on tactile switch
X _{CRYST}	- 11.0592MHz crystal
PZ1	- Piezo-electric buzzer
LCD module	- 16×2 line LCD

switches. The internal timers of the microcontroller are used to provide one-second delay for the digital clock. The hex values fed to timer-1 registers TH1 and TL1 make timer-1 provide a delay of 10ms according to 11.0592MHz crystal. With some internal software logic, one-second delay is formed using 10ms delay of timer-1. Timer-1 is configured in 16-bit mode for the desired operations through TMOD register. Thus current time is updated every second.

The software of the digital clock is interrupt-based. Whenever timer-1 overflows, an interrupt is generated by

the microcontroller. This interrupt is provided by the interrupt service routine at vector 001BH. The jump command written at this vector initiates the `intr_service` routine. All the functioning of the digital clock with alarm is controlled by this service routine.

By default, all the ports act as output. The software makes port pins P2.4 through P2.7 of the microcontroller act as input. The software also handles the functioning of all the switches—up, down, enter and alarm. Pooling and identification of switches and limits for up and down too are provided by the software.

Construction and testing

An actual-size, single-side PCB of the microcontroller-based digital alarm clock is shown in Fig. 3 and its component layout in Fig. 4. Use IC base for microcontroller AT89C51. Also, use a heat-sink with voltage regulator 7805 (IC2) to avoid any damage to the circuit. Check continuity between respective connections using a multimeter.

Initially, by varying preset VR1, set the contrast level for proper display on the LCD. If the reading or display is not steady, check for loose connections or dry soldering joints.

Note. The source code of this article is included in this month's EFY-CD and is also available on www.efymag.com website. ●

An M.Tech in electronic product designing and technology from Punjab Engg College, Chandigarh, in collaboration with CDAC, Mohali, the author is currently working with CPA Global as a patent researcher

THE COMPLETE MAGAZINE ON OPEN SOURCE

LINUX

BETA
ForYou

Your favourite Linux Magazine is
now on the Web, too.

LinuxForU.com

Follow us on Twitter@LinuxForU